Title: Detection of Food Derived Volatile Organic Compounds Using Metal Functionalized Titanium Dioxide Nanotube Arrays

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ABSTRACT:

There is a real need for advanced sensing technology to address significant risks to the food supply. By 2050, growing populations and aspirations for improved quality of life are projected to increase food demands 70%. Elimination of food wastage due to spoilage or contamination during storage could minimize these demands. One method to address this problem is to use point of use sensors that can be deployed throughout the food supply chain. Undetected microbial contamination (i.e. bacteria/fungus) affect human health and leads to health risks to large populations across the world. Food contaminated with harmful pathogens, microbes or chemical substances is linked to more than 200 diseases. Unsafe food can be linked to an estimated 2 million deaths annually, including children. Often the food contamination occurs at the source and point-of-use rapid detection could enable timely intervention and minimize damage and wastage. This presentation focuses on novel material systems based on functionalized titanium dioxide arrays for rapid detection of food pathogens based on the presence of volatile organic compounds given off by biological pathogens present in the food.